POSIX storage requirements

Testable requirements for working with files held on connected POSIX machines through the portal. Each one serves part of the INN Digital Infrastructure Strategy (2026). The access rules themselves are in access-control.md and apply here unchanged; this register holds only what is specific to POSIX machines. How each one is met, and which ones need administrator access on a machine, is in the design (docs/posix/README.md).

Open items name the question that settles them in *INN use cases: decisions* (27 October 2026): A1 to A3 in Part A, D1 to D9 in Part D.

Status: *settled* (agreed; tests required), *proposed* (awaiting a functional decision), *open* (needed by the strategy, not yet decided; no tests yet).

ID	Requirement	Serves (strategy)	Status
PX-1	Through the portal, a user can browse, download and upload files held on a connected POSIX machine.	4.1 steps 2–4	settled
PX-2	Through the portal, a user can delete files held on a connected POSIX machine.	4.1	proposed
PX-3	A request gets the same answer on a POSIX machine as on object storage: the same access rules, refusals and listings.	4.1; 5.1 postcondition	settled
PX-4	Identifiable data considered so far is held on the INN and ImageTech machines and scrubbed there; only scrubbed data leaves them. What the portal must do to support this: D3.	5.1; 4.7	open
PX-5	An interrupted download resumes rather than restarts.	5.8 main flow step 3; 4.1 outcome (large files)	proposed
PX-6	An upload is either complete or absent: a failed upload never leaves part of a file, and never replaces an existing file.	5.1 postcondition	proposed
PX-7	File contents move directly between the user and the machine, not through the portal's servers.	5.8 main flow step 2	proposed
PX-8	It works on a machine where INN has no administrator access. Where INN has it, it is used only where another requirement needs it.	decisions Part A (the setup balance)	proposed
PX-9	Who decides who else can open the governed files on each machine is as A1 decides.	5.1 postcondition	open
PX-10	Activity on POSIX data is recorded as A2 decides (access through the portal is also covered by AC-9).	4.1 outcome (transparent logging); 5.1 non-functional requirements	open
PX-11	Files created, changed or deleted directly on a machine appear in the portal within the time A3 sets.	4.1 step 2	open
PX-12	Identifiable data moves between the INN and ImageTech machines as D1 decides.	5.1; 4.7	open

PX-1 and PX-3 come from the decisions document's settled list. PX-4's scope is settled there too, but what the portal must do about it waits on D3, so it has no tests yet. PX-8 states the aim of the setup balance in Part A and awaits that decision. The answers to A1 to A3 decide which requirements need administrator access; the design lists them.